Fall 2026 Project Lab – Financial Mathematics Program

Company:

Northern Trust Asset Management

50 S Lasalle St., Chicago, IL
https://ntam.northerntrust.com/united-states/institutional/insights/research

Team Size: up to 6 students

Company representatives:
Manan Mehta (Booth alumnus) – Head of Quant Fixed Income Strategies
Paul Stevenson (MSFM alumnus 2016) – Quantitative Fixed Income Portfolio Manager
Edmund Wadge - Quant Research, Senior Associate

Meeting format:
Virtual meetings – team will meet at least biweekly.

Non-disclosure agreement:
Yes, required for this project

Project Topic:
LLM Based Earnings Surprise

Background & Objective:
The reaction of a stock's price to its earnings announcement is among the most-studied and least-explained events in empirical finance. Fifty years of accounting research raised the explained variation(R2) of announcement-window returns from roughly 2%(Lev, 1989) to about 4-5% (Chiang et al., 2019) and using dozens of hand-constructed surprise measures, to roughly 7.5-8% (Hand et al., 2022).

Koijen and Levy (2026) show that frontier LLMs reading the earnings-call transcript raise this to 14% and 17% with gradient free prompt/context optimization(ACE). LLMs have improved to the point where they can systematically extract information that is economically important but difficult to capture in structured datasets. The opportunity is not in headline metrics. It is in the judgments, assumptions, and disclosures that explain why the numbers look the way they do.

Consider a change in accounting estimates disclosed in a 10-K of Google. Extending the useful life of servers can add billions to reported earnings without changing the underlying cash flows. The earnings impact is easy to observe; the real signal is management's rationale and what it reveals about the business. In isolation, these disclosures are company-specific. Across a peer group, they become industry signals. When multiple hyperscalers extend server lives within a short period, the question shifts from accounting to economics: are assets genuinely becoming more productive, or are reporting assumptions changing? That type of context is difficult to capture in traditional factor models but is increasingly accessible through LLMbased analysis. The hypothesis is that these narrative disclosures contain incremental information on future earnings outcomes and can be transformed into targeted alpha signals or screening tools within the research platform.

There is also plenty of evidence in the benefits of separating recurring and non-recurring components of net income to forecasts future performance For example: Rouen, So, Wang Core Earnings New Data and

Evidence.pdf . This proposal is primarily based on Koijen, Ralph S. J. and Levy, Bradford (Lynch), Assessing the Benefits of Optimized Agentic AI Systems for Asset Pricing (June 30, 2026).

Skills required:
Python will suffice; access to Latest LLM models; GPUs for Inference; Factiva for Earnings Transcripts

Project objectives or questions to be investigated:
This proposal seeks to develop a more robust estimate of the surprise element in an earnings release using large language models and fundamental data. There is long standing literature on surprise unexpected earnings

Develop a point-in-time, security-level measure of total earnings-announcement surprise:

The measure should capture:
1. Headline surprise: EPS actual versus analyst expectation.
2. Fundamental-vector surprise: revenue, margins, cash flow, segments, KPIs and other reported line items versus expectations.
3. Forward surprise: new or revised management guidance versus prevailing forecasts.
4. Narrative surprise: semantic changes in explanations, risks, confidence, causal attribution and Q&A relative to prior disclosure.
5. Persistence/quality: whether items in financial statements are recurring, transitory, accounting-driven or operational.
Uncertainty: forecast dispersion, disagreement, confidence and missing-information penalties.

Implementation details:
See following pages.

Data/Computing Resources:
Students should expect to use University-provided data sources for this project.

High performance/scalable computing is required for this project which will be provided by the University. Google colab with GPU access should suffice but the project does require the ability to invoke LLMs via CLI.

Project Lab Policy:
Project Lab is a research course (FINM 36000/01). There is absolutely no employment relationship between project sponsors and students.

Students may add this project lab to their resume, including the company name and a general description of their research (abiding by any applicable NDA).

Students should use the following format:

[Company Name]	Chicago, IL
Quantitative Research – University of Chicago Project Lab	October 2026 – Present

LLM Based Earnings Surprise

Submitted by: Manan Mehta

Date: 08/21/2026

Strategy area: ☒ Alpha signals ☐ Alpha modelling ☐ Responsible investing
☐ Client PM ☒ Portfolio construction ☐ Product development

Products: ☐ MF ☒ AE ☐ QDF ☐ QESG ☐ QLV ☐ QSCC ☐ 'Local alpha' (i.e. PIF, ASX)
☐ Quant-SI solutions ☒ QFI ☐ Trading

1. Goal

This proposal seeks to develop a more robust estimate of the surprise element in an earnings release using large language models and fundamental data. There is long standing literature on surprise unexpected earnings

Develop a point-in-time, security-level measure of total earnings-announcement surprise:

$$\text{LLM-SUE}_{i,t} = f\left(\underbrace{\text{new release/call information}_{i,t}}_{\text{what was learned}}, \underbrace{\Omega_{i,t^-}}_{\text{what investors plausibly expected}}\right)$$

The measure should capture:

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2. Business value

This will further our research in the areas of NLP leveraging LLMs, and the value of leveraging this approach has been validated by several external research publications and also a live competition

forecasts that explicitly enforce a knowledge cutoff. This can be extended to and applied to quant credit strategies as well.

3. Ultimate use case

This will be a short-term alpha signal that can be incorporated as a screen or as a targeted alpha factor.

4. Economic motivation

The reaction of a stock's price to its earnings announcement is among the most-studied and least-explained events in empirical finance. Fifty years of accounting research raised the explained variation(R2) of announcement-window returns from roughly 2%(Lev, 1989) to about 4-5% (Chiang et al., 2019) and using dozens of hand-constructed surprise measures, to roughly 7.5-8% (Hand et al., 2022).

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5. Data

Need access to the full text of earnings releases and multiple enterprise approved LLMs. The token cost of making such calls needs to be considered.

6. Approach

Based on a working prototype of this seeing R2s in the range of 15-20%. The AI system can explain cross-sectional variation in stock returns around earnings announcements (CAR) using only information available at the time of the announcement, versus about 5% using a traditional earnings-surprise measure and about 8% in the prior literature.

Step 1: Establish Prior Expectations

What does the market already know?

- Analyst estimates and consensus expectations
- Historical financials and trends
- Prior company disclosures and guidance
- Industry and peer context
- Current valuation/spread levels

Step 2: Ingest New Information

What did management just tell us?

- 10-K, 10-Q, earnings call, investor day
- Accounting policy changes
- Risk factor updates
- Capital allocation decisions
- Operational and industry commentary
- Segment-specific disclosures

Step 3: Identify the Delta

What changed relative to prior beliefs?

- New disclosures versus prior filings
- Changes in management language
- Changes in assumptions or estimates
- New risks, opportunities, or strategic priorities
- What management is emphasizing more or less

Step 4: Convert into Structured Signals

Use LLMs to classify and quantify the information.

- Management confidence

- Accounting aggressiveness/conservatism
- Investment intensity
- Competitive positioning
- Industry outlook
- Disclosure novelty and persistence

Step 5: Cross-Sectional Context

Is this firm-specific or industry-wide?

- Compare disclosures across peers
- Identify emerging industry themes
- Separate company effects from sector effects
- Measure uniqueness versus consensus behavior

Step 6: Alpha Signal Construction

Transform insights into investable features.

- Ranked score that is targeted at predicting a longer term

Step 7: Validation

Does the signal predict future outcomes?

- Excess returns
- Spread tightening/widening
- Rating actions
- Earnings revisions
- Fundamental deterioration/improvement

7. Workload

Meaningful – but can be outsourced to another team.

8. Project stage-gates and evaluation criteria

Stage gate	Description/evaluation criteria
1. Long short	- Literature review - Economic rationale - Data quality - Initial signal construction

	- Long short results with a focus on cumulative abnormal returns over 1 week, 4 week, 12 week horizons. - Prospect of added value (i.e. attribution, implementation)
2. Long only	- Final signal construction - Implementation in strategy - Realistic strategy simulation (long-only)

9. Literature Review

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Fama, E. F., Fisher, L., Jensen, M. C., & Roll, R. (1969). *The adjustment of stock prices to new information*. *International Economic Review, 10*(1), 1-21.

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Hoberg, G., & Phillips, G. (2016). *Text-based network industries and endogenous product differentiation*. *Journal of Political Economy, 124*(5), 1423-1465.

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